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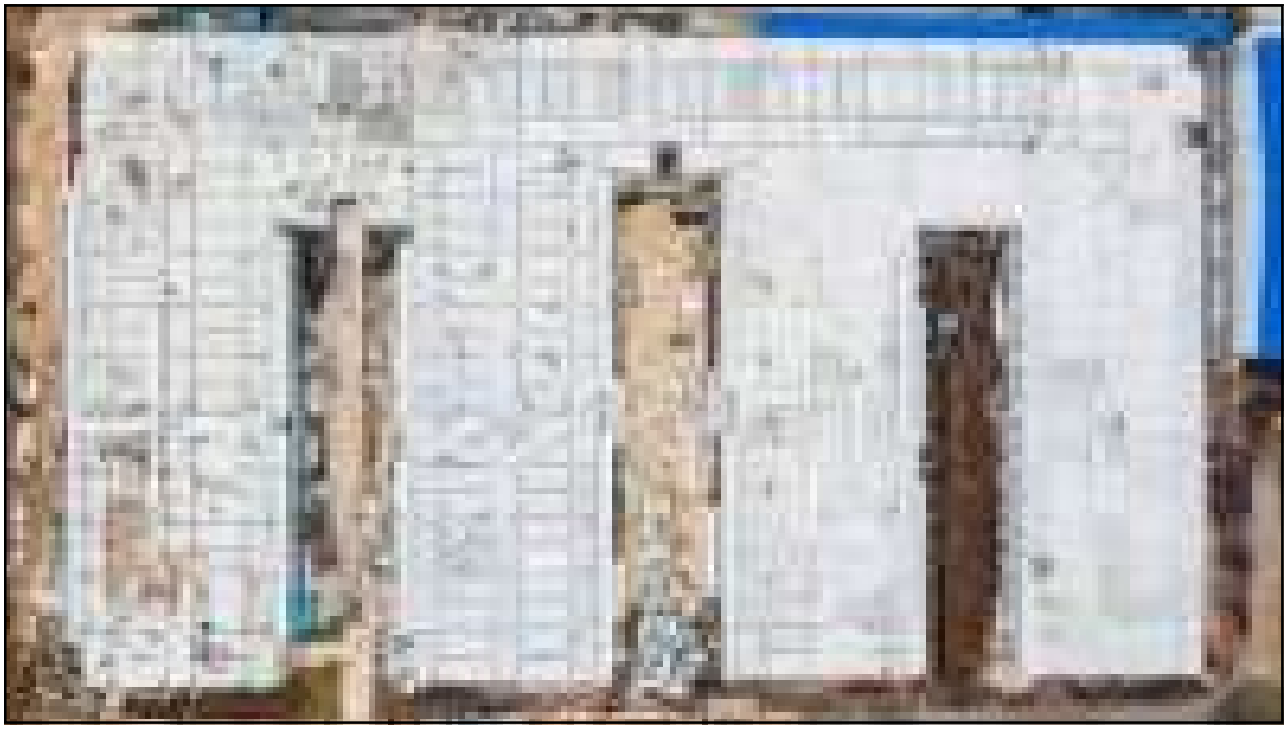
香 港 大 學
THE UNIVERSITY OF HONG KONG

CIVL2108 Principles of Civil Engineering Management

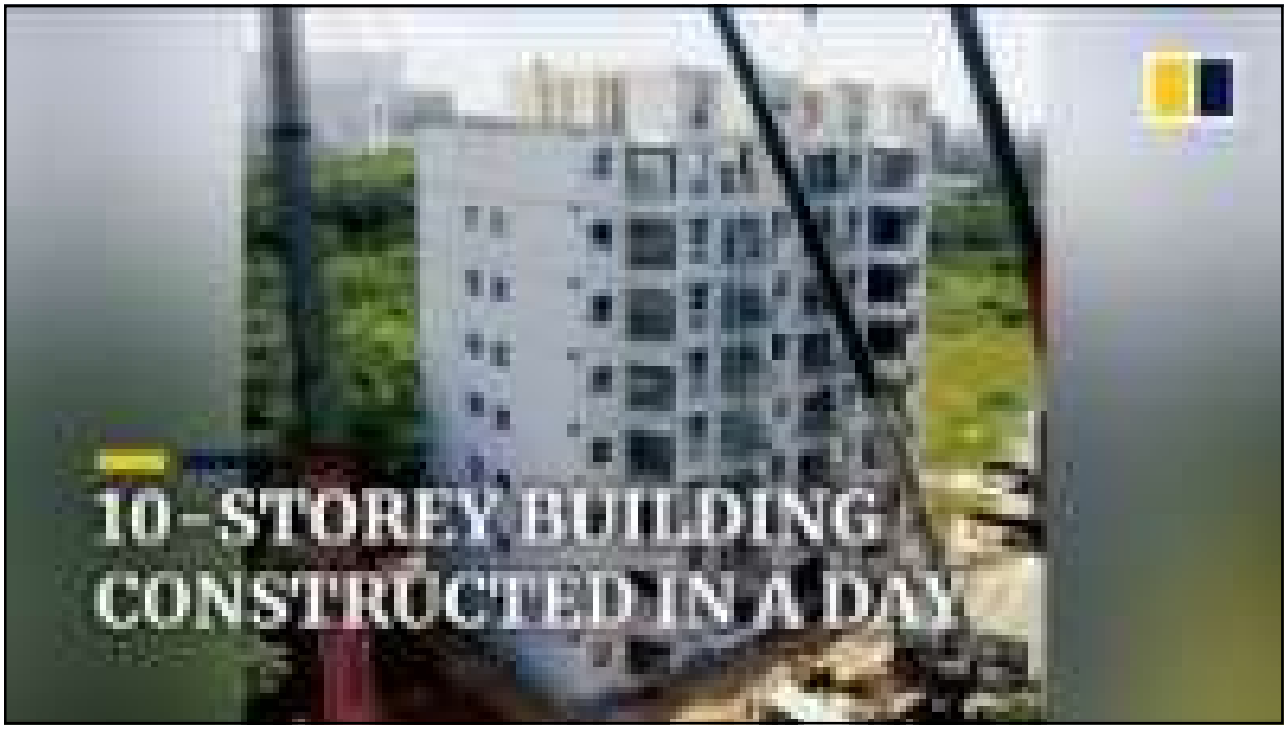
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1. Construction Industry- overview

- **Societal relevance**
  - One of the first businesses that humankind developed
  - Virtually all other businesses rely on the construction industry to provide and maintain their accommodation, plants and infrastructure
  - Determinant of where and how almost everyone lives, works and plays





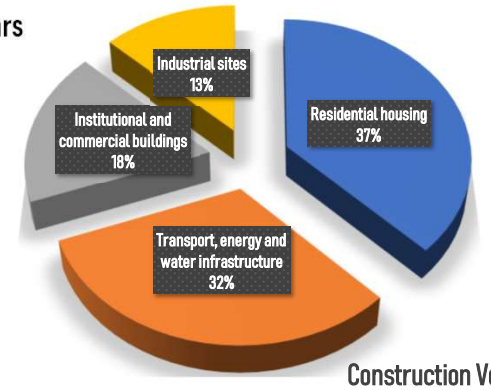


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1. Construction Industry- overview

- **Economic relevance**
  - **6%** of global GDP (5% in the developed countries, >8% in developing countries)
  - Employment > **100 million people**
  - 1% GDP investment, 1.5% increase in GDP after 4 years



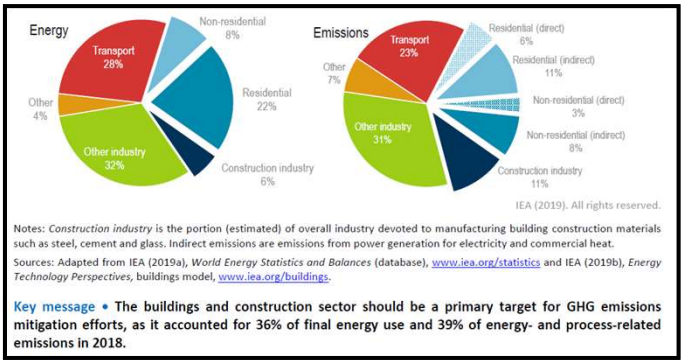
Construction Volume

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1. Construction Industry- overview

- **Environmental relevance**
  - Single **largest** global consumer of resources and raw materials
  - **50%** of global steel production
  - **40%** of solid waste (in the US)
  - **25-40%** of the world's total carbon emissions



Notes: *Construction industry* is the portion (estimated) of overall industry devoted to manufacturing building construction materials such as steel, cement and glass. Indirect emissions are emissions from power generation for electricity and commercial heat.

Sources: Adapted from IEA (2019a), *World Energy Statistics and Balances* (database), www.iea.org/statistics and IEA (2019b), *Energy Technology Perspectives*, buildings model, www.iea.org/buildings.

Key message • **The buildings and construction sector should be a primary target for GHG emissions mitigation efforts, as it accounted for 36% of final energy use and 39% of energy- and process-related emissions in 2018.**

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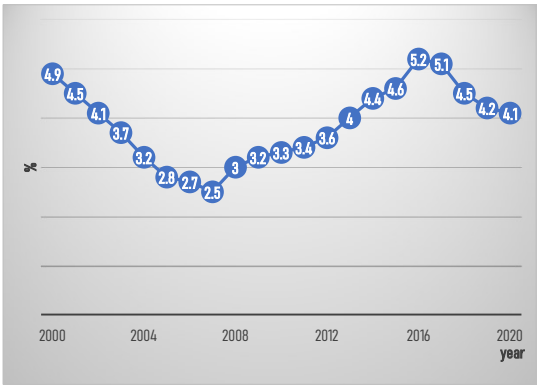
1. Construction Industry- overview

❖ Hong Kong Chapter

Hong Kong GDP from Construction (2000-2022) (In HKD Million)
Average 22463.90 HKD Million



GDP by Major Economic Activity (%)

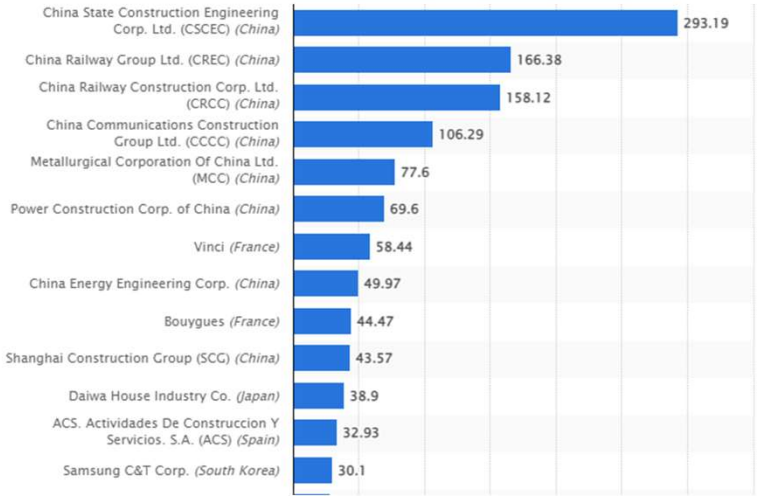


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1. Construction Industry- overview

❖ Largest construction firms based on revenue in the world in 2021



Source: statista

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1. Construction Industry- overview

◆ WSJ NEWS EXCLUSIVE | PROPERTY REPORT

Warren Buffett to Offer a New Spin on Modular Construction

A startup owned by Berkshire Hathaway aims to make the construction industry more like car manufacturing

REAL ESTATE | PROPERTY REPORT

Startup Backed by SoftBank Pours Money Into a Bid to Reshape Construction

Katerra, tech-focused maker of modular building parts, raises over $1 billion, but questions remain over profitability

REAL ESTATE | PROPERTY REPORT

How a SoftBank-Backed Construction Startup Burned Through $3 Billion

Katerra's downfall shows how Silicon Valley's strategy of growth at any cost can backfire in complicated industries like real estate

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2. Major Types of Construction

- By trade group

I. Construction works at construction sites
   - Private sector construction sites
   - Public sector construction sites
   - Pubic Private Partnership (PPP) construction sites

II. Construction works at locations other than sites
   - General trades

Decoration, repair and maintenance, and construction works at minor work locations such as site investigation, demolition and construction alteration and additional works

   - Special trades

Carpentry, electrical equipment, ventilation, gas and water fitting installation and maintenance

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2. Major Types of Construction

- By end-use group

I. Buildings
  - Residential
  - Commercial
  - Industrial & Storage
  - Service

II. Structures & Facilities
  - Transport
  - Other utilities & plant
  - Environment
  - Sports & recreation

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2. Major Types of Construction

- By nature of construction activity

I. Site formation & clearance
II. Piling & related foundation works
III. Erection of architectural superstructure
IV. Civil engineering construction









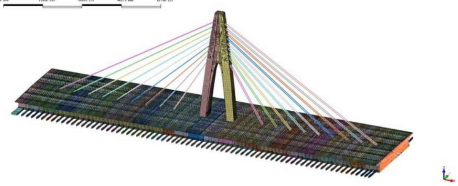
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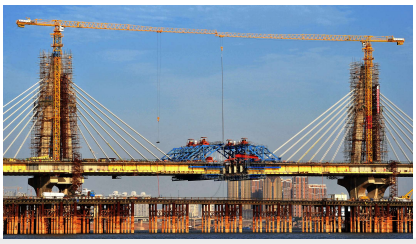
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3. Project life cycle

- Market Demands or Perceived Needs
- Conceptual Planning and Feasibility Study
- Design and Engineering
- Procurement and Construction







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3. Project life cycle – Cradle to Grave

- Startup for Occupancy
- Operation and Maintenance
- Disposal of Facility



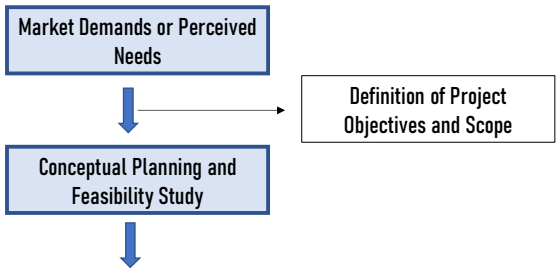




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3. Project life cycle



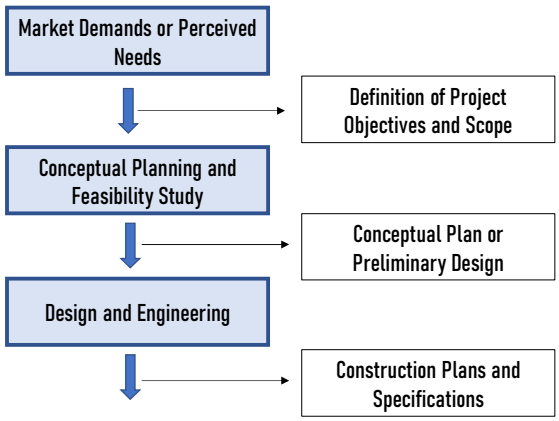
Conceptual Planning and Feasibility Study

- How does this project fit with stakeholder goals, strategies, and needs?
- How will this project be paid for?
- How will the public and other stakeholders react?
- What are the political ramifications?
- Can it be done?
- What are the risks
- What might it look like?
- How can the project be executed?
- Best approaches?
- Alternatives?
- Risks and opportunities?
- Design options?
- Can the project be constructed?

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3. Project life cycle



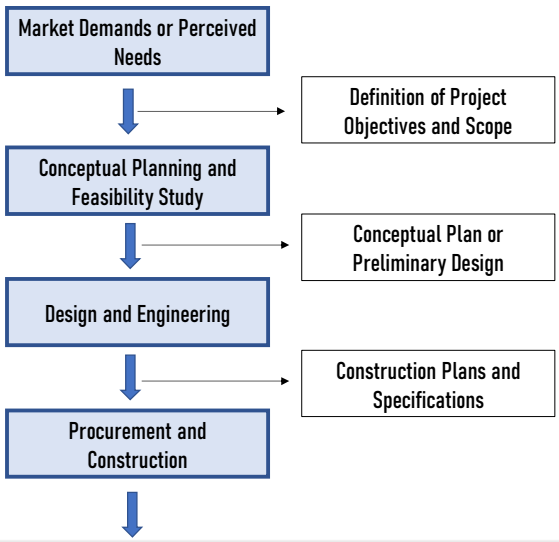
Engineering & Design

- Architectural concept
- Evaluation of technological process alternatives
- What are the final design parameters? What are the key loadings
- How will key elements and features be designed, sized, tested
- What materials will be used? What equipment will be specified (capacity and sizing)?
- What are the phases of construction?
- How are the design choices impacting cost and schedule?
- Comparative economic studies

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3. Project life cycle



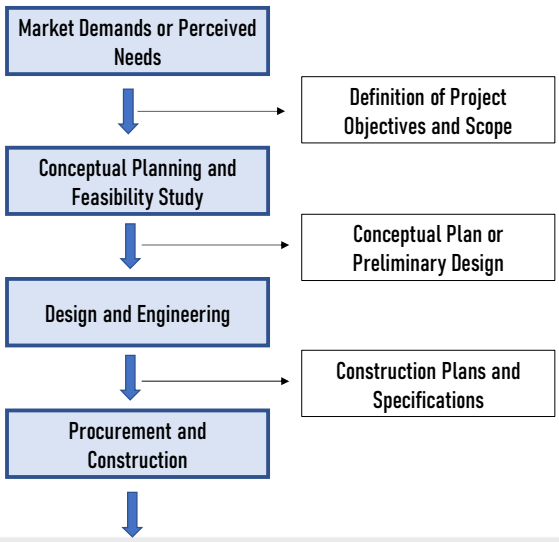
Construction

- How will it be built?
- What are the phases of construction?
- How will the risks be mitigated?
- How will safety be carried out?
- What equipment and resources will be required?
- What materials are required?
- How will the equipment and materials be procured and delivered timely?
- How will the subcontractors be coordinated?

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3. Project life cycle



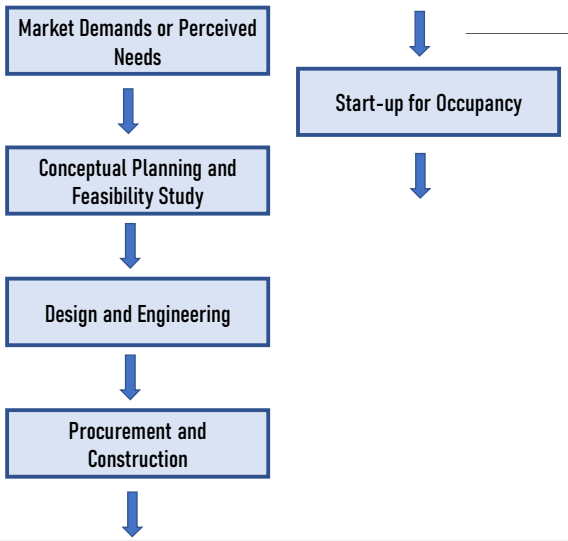
Procurement

- How do we adequately describe the scope and intent of the project?
- Which contractor/team is best suited to build the project?
- What is their experience?
- What is their understanding of the risks and key project elements?
- Do they have the leadership and resources?
- Do they have the capacity?
- Do they have the ability to meet the schedule?
- What is their commitment to safety?
- What is their commitment to the environment?

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3. Project life cycle



- Start-up for occupancy

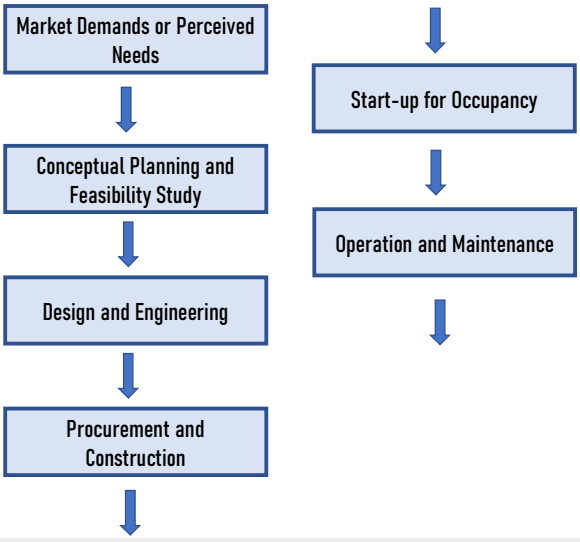
Ensure the facility has been completed

- Test facilities before and during the initial start-up period-function as a total system
- Adjustment or correction may be needed to achieve optimum performance

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3. Project life cycle

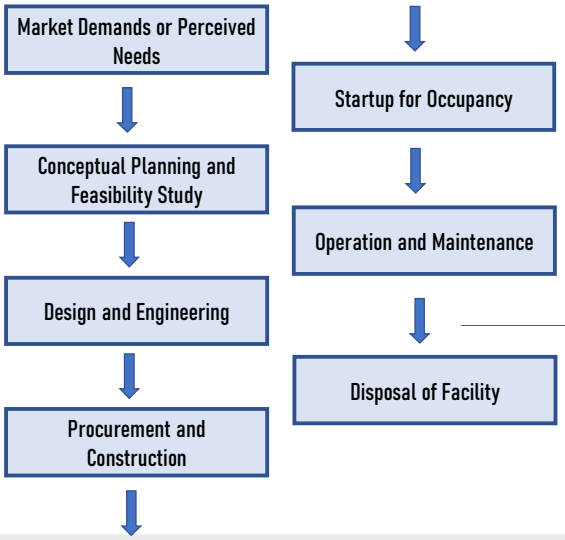


- Operation and maintenance
  - How will the constructed project be efficiently and effectively used?
  - How will the asset be maintained to ensure its design life?
  - Can the asset be improved?

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3. Project life cycle



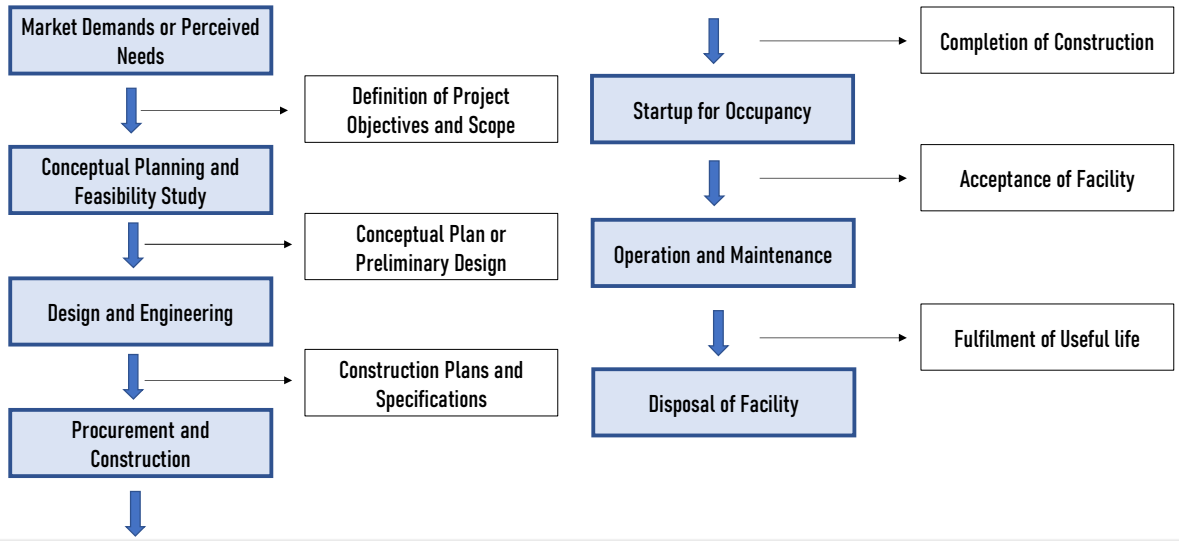
Disposal of facility

- What is the current status and need for the asset?
- What is the state of repair?
- What are the alternatives such as refurbish, alternative use, dismantle, etc.?
- How can the materials/components be reused, recycled, or safely disposed of?

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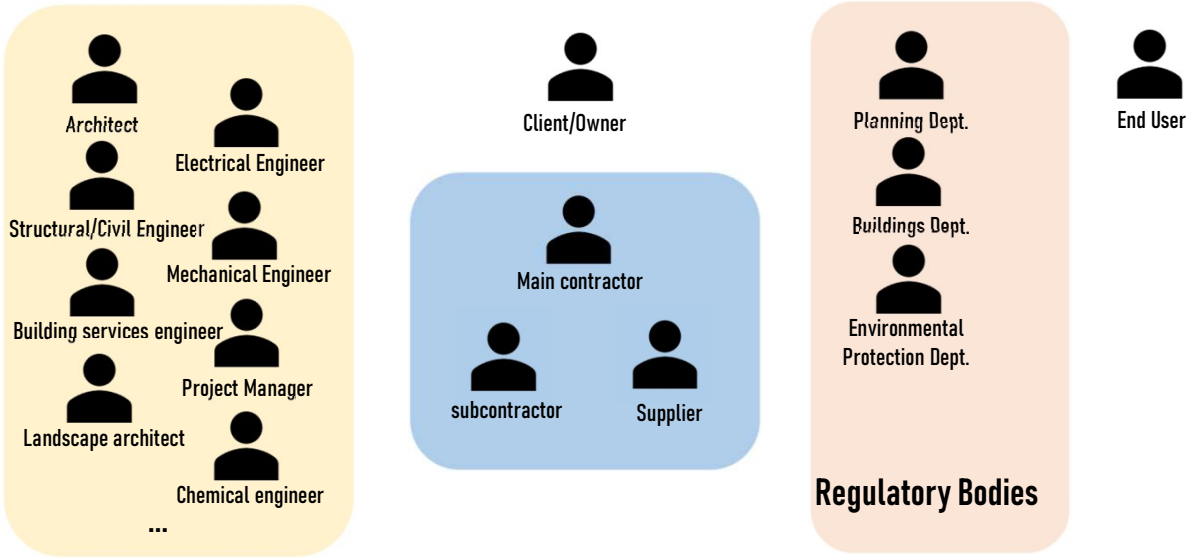
3. Project life cycle



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4. Project participants



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4. Project participants



Private: Individual, corporation or consortium

Public: government departments, public authorities, etc.

Quasi-governmental: Hong Kong Airport Authority, Urban Renewal Authority, etc.

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4. Project participants

Client/Owner

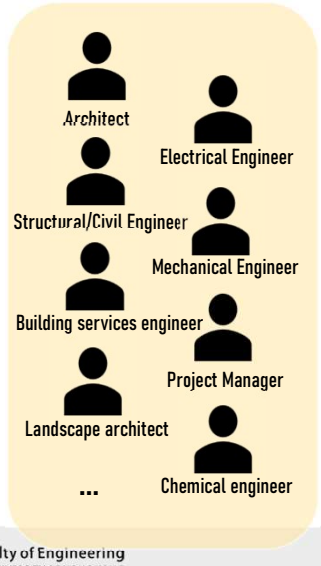
- Fulfilment of contractual obligations to other team members, including furnishing site and related information, and timely payment
- Compliance with applicable laws, regulations, codes, standards, and practices
- Provision of adequate funding
- Provision of necessary real estate or right(s) of way
- Provision of project quality assurance oversight to verify established goals and objectives
- Fulfilment of insurance and legal requirements
- Assignment of site safety responsibility
- Acceptance of completed project

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4. Project participants

Design professionals



Architect – functional and aesthetical aspects

Structural/civil engineer – structural aspects

Building services engineer (BSE) – lift, fire, electrical, plumbing and drainage, gas, control system, etc.

Landscape architect – external works

Quantity surveyor (QS) – cost and contractual aspects

Project manager (PM) – coordination, monitoring project's time, cost & quality, etc.

Financial advisor – economic & financial viability, tax law affecting investment decision

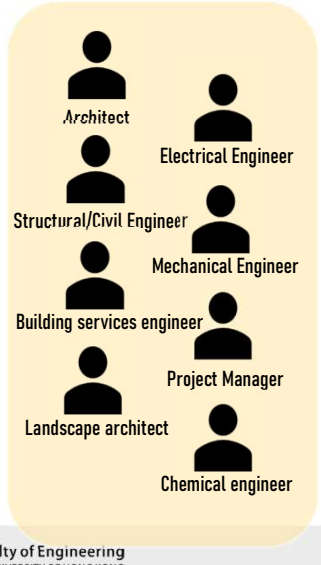
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4. Project participants

Design professionals

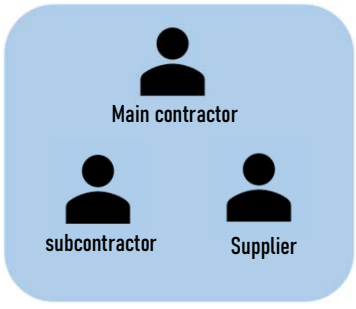


- Fulfilment of contractual obligations to other team members
- Compliance with applicable laws, regulations, codes, standards, and practices
- Fulfilment of professional standard
- Development and drafting of well-defined project contract documents
- Responsiveness to project schedule, budget and program
- Provision of construction-phase design services

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4. Project participants



Main contractor (MC) – directly responsible for the whole project

Domestic subcontractor (DSC) – selected by MC to perform part of a project, e.g., concreting

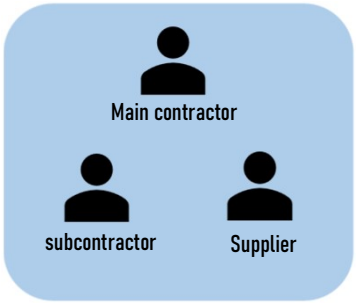
Nominated subcontractor (NSC) – selected by client & appointed by MC to perform a specialist task

Nominated supplier (NS) – selected by client & appointed by MC to supply specific material(s)

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4. Project participants



- Fulfilment of contractual obligations to other team members
- Compliance with applicable laws, regulations, codes, standards and practices
- Interpretation of project drawings and specifications
- Construction of facility as described in contract documents
- Management of construction site activities and safety program
- Management, quality control and payment of subcontractors and vendors

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4. Project participants



Planning and design

General public – public consultation
LegCo – budget approval for major public schemes
Government departments
E.g. Planning Dept., Buildings Dept. and Environmental Protection Dept.

Regulatory and control
Inspect the completed works
E.g. Buildings Dept., Fire Services Dept., Water Supplies Dept., China Light, Town Gas, etc.

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4. Project participants

End User

Operating staff – e.g. doctors operating a hospital

Beneficiaries – e.g. patients using a hospital

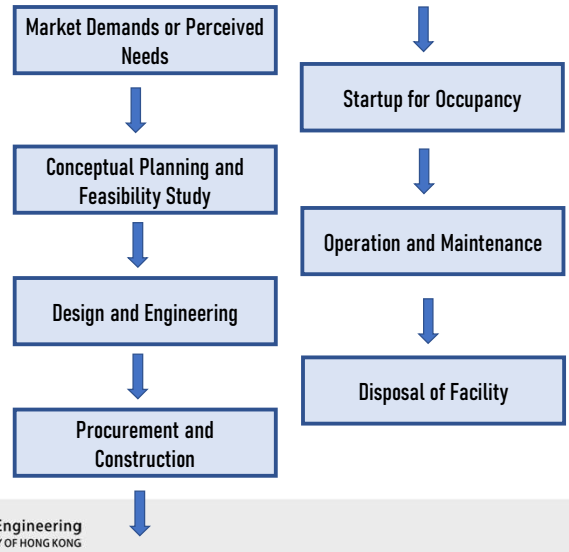
Users' needs vs. client's requirements

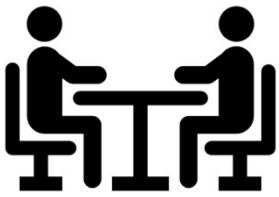
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4. Project participants

- List the participants who are likely to be involved in the following stages of a private residential scheme:





Discussion: 10mins

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4. Project participants

Market Demands or Perceived Needs

Client, financial consultant, market analyst & general practice surveyor

Conceptual Planning and Feasibility Study

- Project manager (PM) to be appointed by the client
  - Set up work plan & schedule
  - Appoint consultants (design team) & specialists
  - Help establish client & user requirements
- Design team — Architect, structural/civil engineer, BSE – deal with design aspects & conduct preliminary investigations; Quantity surveyor – prepare a preliminary cost estimate
- General public – provide opinions during public inquiry stage
- User representatives (if necessary) – advice on user requirements

Design and Engineering

- Client's PM – coordinate the design team & prepare preliminary programme
- Architect, structural /civil engineer, Building Services Engineer (BSE) – prepare drawings, schedules & specifications
- Quantity surveyor (QS) – produce final cost estimate
- Regulatory bodies – approve the design
- General public – reflect opinions during gazetting stage
- LegCo – approve funding for public schemes

Public work

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4. Project participants

Procurement

- MC – price the tender, obtain tenders from domestic subcontractors (DSC) and suppliers, submit tender
- QS – evaluate tender & prepare tender report
- Architect, structural /civil engineer, BSE– evaluate technical aspects of tenders
- Client's PM – overall coordination

Construction

- Contractor's PM – establish programme, organisation, resources, construction method & sequence
  Supervise works of DSC, suppliers, NSC, NS
- Client's PM – monitor contractor's conformance
- Architect, structural / civil engineer, BSE – issue variations & certify payment
- QS – monitor client and contractor's budget as a result of variations, prepare interim payments (IP)

Start-up for Occupancy

- Regulatory bodies – test the facilities according to statutory requirements
- Client's PM & design team – check for acceptability
- Contractors –start-up, test and adjust facilities
  Prepare records – produce as-built drawings
  Prepare operating instructions & train users

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4. Project participants

Operation and Maintenance

- Client or user – identify defects
- Contractors – rectify defects
- QS – prepare final payment (final account)

Disposal of Facility

- Owner and user – find out the efficiency of facility
- Economist – determine the value for selling the facility